
Post-Cyclone Displacement Spike Prediction in the Indian Sundarbans

*A Block-Level Machine Learning Framework
for Anticipatory Displacement Intelligence*

[ABHIRUP HALDER]

[Undergraduate independent Researcher]

[Kolkata, India]

halderabhirup3101@email.com

Keywords: Cyclone displacement prediction · XGBoost · SHAP
interpretability · Sundarbans · Climate vulnerability · Temporal validation ·
Disaster risk reduction

Study Region: Indian Sundarbans, West Bengal, India

Cyclones: Aila (2009) · Amphan (2020)

Spatial Unit: Block level (n = 19)

Model: XGBoost Gradient Boosting Classifier

Accuracy: 84.2% · Macro AUC-ROC: 0.906

Date: May 26, 2026

Abstract

The Indian Sundarbans — home to 4.5 million people across 19 permanently inhabited blocks in South 24 Parganas, West Bengal — is one of the most cyclone-exposed deltaic regions on earth. Following every major cyclone, displacement is not random: the same blocks empty, the same embankments breach, and the same structurally fragile households bear a disproportionate share of the humanitarian burden. Yet disaster management systems across the region remain reactive by design, activated after landfall rather than before it.

This study develops the first block-level displacement spike prediction framework for the Indian Sundarbans, combining a two-scale XGBoost gradient boosting architecture with SHAP (SHapley Additive exPlanations) interpretability analysis. A macro-scale model incorporating event-specific meteorological variables characterises overall storm severity, while a micro-scale structural model isolates block-level vulnerability features — poverty, elevation, flood inundation extent, mangrove cover, embankment breach history, and housing quality — independent of storm magnitude. The structural model is trained on Cyclone Aila (2009) and validated out-of-sample against Cyclone Amphan (2020), achieving 84.2% test-set accuracy and a macro AUC-ROC of 0.906.

The central finding — that structural vulnerability predicts displacement intensity more reliably than meteorological magnitude, with the meteorological model collapsing to 26.3% accuracy against the structural model’s 84.2% — directly challenges the hazard-centric architecture of existing early warning systems and confirms that block-level displacement is a structurally predictable phenomenon amenable to anticipatory resource allocation. SHAP analysis identifies flood inundation extent and mean elevation as the dominant predictors, with mangrove cover contributing a measurable protective signal consistent with the ecological buffering literature. Policy implications for the West Bengal Disaster Management Authority and comparable institutions across the Bay of Bengal coast are developed.

JEL Classification: Q54 · R12 · C53

Word Count: Approximately 12,000 words

Contents

1	Introduction	4
2	Study Area	5
2.1	Geographic Setting	5
2.2	Ecological Character	6
2.3	Socio-economic Profile	6
2.4	Cyclone Exposure and Displacement History	6
3	Literature Review	7
3.1	Theoretical Foundations: Why People Move	8
3.2	The Scale of the Problem: Global Displacement Projections	8
3.3	Social Vulnerability and Structural Drivers	9
3.4	The Timing and Mechanism of Displacement	9
3.5	Climate, Economics, and Quantitative Methods	10
3.6	The Gap This Study Fills	10
4	Methodology	11
4.1	Research Design and Temporal Validation Framework	11
4.2	Data Sources	11
4.3	Feature Engineering	12
4.4	Model Architecture	12
4.5	The Nine Features	13
4.6	Why XGBoost	14
4.7	How the Model is Configured	15
4.8	The Ablation Test	15
5	Results	15
5.1	Model Performance: Two-Scale Architecture	15
5.2	Block-Level Predictions	17
5.3	SHAP Feature Importance	18
6	Discussion	19
6.1	Structural Vulnerability Dominates Hazard Magnitude as a Displacement Driver	19
6.2	Feature Dominance and Model Interpretation	20
6.3	The Mangrove Signal	21
7	Policy Implications	21

8	Limitations	22
8.1	Sample Size and Statistical Power	22
8.2	Feature Dominance	22
8.3	Static Feature Engineering	22
8.4	Displacement Outcome Variable	22
9	Conclusion	22

1. Introduction

The Sundarbans – A mangrove so mysterious and dangerous that it earned a nickname: THE LAND OF EIGHTEEN TIDES, notoriously known as “The Realm of Dakshin Rai”, the lord of the South, the deity that governs what lives and what dies in the forest. Well, now you will be thinking why such an abstract name for such a lovely forest. Well, in nature, the more dangerous the thing, the more beautiful it seems. The main reason for its mystery is that you would not know when the danger is about to strike because it’s always a surprise. This jungle is (10,000km²) in size, not just the largest Mangrove Ecosystem in Asia, but globally also. While conceptually smaller than the Amazon River Delta (38,304 km²), it is distinct from the Amazon: a maze of massive halophytic mangrove forests filled with animals waiting to take your life. It is a deltaic archipelago where even the sturdy islands of Sundarbans had to bow down against the terror of the sea, disappearing and reappearing twice a day like the ‘Marooner’s Rock’ from Neverland in Peter Pan or like the ‘Isla de Muerta’ from Pirates of the Caribbean. It is home to almost 4.5 million people who mostly work as fishermen and honey collectors. The worst part of this forest is that everything is ready to kill you: from man-eating, mysterious Royal Bengal tigers to thirsty Estuarine Crocodiles patrolling its brackish channels; this difference makes it different from the Amazon because it is a rainforest distributed across a continent, while this is a mangrove that is dense, tidal, alive, mysterious and silently hostile.

But the Sundarbans we know today face a threat no animal or reptile prepared its people for. The Bay of Bengal is getting warmed up. Cyclones are becoming destructive and dangerous, and when they arrive, they become the “Dealers of Thunder and Death”- as Cyclone Aila did in May 2009, and Cyclone Amphan in May 2020 – they didn’t just merely damage the Sundarbans. They empty it. Entire blocks of 200,000 to 400,000 people can be displaced in a matter of hours, their embankments breached, their paddy fields swallowed by saltwater, their homes reduced to the mud they were built from. The IDMC (Internal Displacement Monitoring Centre) recorded over 500,000 people displaced by Amphan in West Bengal alone — and that number captures only those who reached official relief camps. What makes this crisis particularly difficult is not the cyclones themselves. It is the response to them. Every time a major storm strikes the Sundarbans, disaster management authorities scramble to understand, in the hours and days after landfall, where the worst displacement has occurred and where resources need to go. The system is reactive by design — built to respond to damage already done, not to anticipate where damage will be worst. And in a region where hours matter — where a family without shelter in the monsoon season faces compounding health, economic, and psychological crises — reactive is not good enough. The deeper problem is that displacement in the Sundarbans is not random. It never has been. Some blocks are empty after every major cyclone. Others, facing the same storm, remain largely intact. The difference is not always

the intensity of the wind or the height of the surge — it is the structure underneath: how low the land sits, how weak the embankments are, how poor the households are, how much mangrove forest remains between the village and the open Sea. This structural pattern — consistent, measurable, and largely ignored by existing early warning systems — is the foundation of this study. This paper asks a direct question to all: “Can we predict, before a cyclone makes landfall, which Sundarbans blocks will experience the highest displacement?” Well, that answer is yes- using an XGBoost gradient boosting classifier trained on block-level geographical, meteorological, and socioeconomic data from Cyclone Aila (2009) and validated against Cyclone Amphan (2020), this study demonstrates with 84 per cent accuracy and a macro AUC-ROC of 0.906. SHAP interpretability analysis further reveals which factors drive that displacement, producing not just a prediction but an explanation — the type of evidence that disaster management authorities, infrastructure planners, and climate policy bodies can actually work with. The contribution of this study is threefold. First, it develops the first block-level displacement spike prediction framework for the Indian Sundarbans, filling a gap in both the climate-migration literature and the applied disaster risk literature. Second, it demonstrates that structural vulnerability at the block level — poverty, elevation, flood inundation extent — predicts displacement intensity more reliably than cyclone hazard magnitude alone, a finding with direct implications for how early warning systems are designed and resourced. Third, it proposes a scalable framework — one built to extend from the Sundarbans to the broader Bay of Bengal coast and eventually to cyclone-exposed deltas globally — positioning block-level displacement intelligence as a replicable tool for climate adaptation policy across the developing world.

2. Study Area

2.1. Geographic Setting

The Indian Sundarbans constitutes the northern apex of the Bay of Bengal delta system, forming the largest contiguous mangrove ecosystem on Earth. Spanning approximately 10,000 km² across 19 blocks in the South 24 Parganas district of West Bengal, the region is geographically defined by a dense network of tidal rivers, estuarine channels, and deltaic islands formed at the confluence of the Ganges, Brahmaputra, and Meghna River systems. The terrain sits at a mean elevation of 3-5 metres above sea level — Hazra et al. (2002) — making it acutely vulnerable to tidal flooding, storm surge, and the compounding effects of accelerated sea level rise. Of the region’s 102 islands, only 54 are permanently inhabited, forming the spatial unit of analysis for this study.

2.2. Ecological Character

The Sundarbans mangrove system is a halophytic forest ecosystem characterised by species such as *Heritiera fomes* (Sundari Tree), *Excoecaria agallocha* (Gewa), *Avicennia alba* (Kalo Baine) and other saltwater plant species adapted to the region's extreme tidal salinity regime. Mukhopadhyay et al. (2015) document sustained ecological degradation driven by upstream freshwater diversion, rising salinity intrusion, and nearly 76 per cent of the area under *Heritiera fomes* (Sundari) declined between 1959 and 2005, with other dominant species, including *Ceriops decandra*, *Excoecaria agallocha*, and *Xylocarpus mekongensis*, also reporting higher rates of decline (Mukhopadhyay et al., 2015). This specific degradation of the tree population has put a lot of anthropogenic pressure on the Delta, with mangrove cover declining by 34.5 per cent between 1975 and 2025. This ecological deterioration is not merely an environmental concern — it is a driver of structural vulnerability, as mangrove density is among the most significant natural buffers against cyclonic storm surge. Pekel et al. document long-term surface water changes across the delta that further constrain the region's natural resilience. Loucks et al. (2010) project that under 28 cm sea level rise trajectories, 96 per cent of remaining mangrove habitat faces inundation by 2070, with direct consequences for the 4.5 million people whose livelihoods and physical safety depend on its buffering capacity.

2.3. Socio-economic Profile

The 54 inhabited islands of the Indian Sundarbans support a population density of 996 persons per km², with an average household size of 4.63 persons, a decadal growth rate of 15 per cent, and a total worker participation rate of 38 per cent, of whom cultivators constitute 4.6 per cent and agricultural labourers 7.4 per cent of the total population (Census of India, 2011). Poverty rates are exceptionally high, with all 19 blocks in the South 24 Parganas, Sundarbans settlements have registered poverty ratios above 30 per cent, and eight blocks recording more than 40 per cent of the population below the poverty line category. Physical infrastructure is severely constrained: the region's 3,500 km of earthen embankments — many structurally deteriorated — constitute the primary defence against tidal flooding and storm surge, with a post-Aila survey identifying 778 km as highly vulnerable and in need of restructuring (Danda et al., 2011) Their structural deterioration represents one of the most consistent predictors of post-cyclone displacement identified in this study.

2.4. Cyclone Exposure and Displacement History

The Sundarbans sits at the top of one of the world's most active tropical cyclone basins. The rate of increase of sea surface temperature over the Bay of Bengal is recorded at 0.019°C per year (1.9°C in a century), resulting in increasingly severe and frequent cyclones

in recent decades. The average elevation of the region is approximately 3 metres above mean sea level, while the average height of storm surges regularly exceeds 6 metres — far above the embankment height of 3.6–5.0 metres — creating the structural conditions for systematic inundation at landfall. So basically, we rely on just 3,500km of mud to defend 4.5 million lives. When a storm surge exceeds 6 metres, these earthen walls don’t just fail; they dissolve, turning the very soil people live on into the engine of their displacement. Sundarbans’ history was written with debris of two catastrophic events: ALIA (2009) and AMPHAN (2020) Cyclone Aila (May 2009) delivered such a devastating kinetic blow that it resulted in at least 100 river embankments being breached by storm surge, with upwards of 2.3 million people displaced as 175,000 homes were destroyed and 270,000 were damaged across West Bengal. Aila damaged or destroyed nearly a quarter of the 3,500 km of mud-based embankments across the Sundarbans. 11 years later, after the terror and trauma of ALIA, the worst cyclone that Sundarbans has ever faced came on 20th of May 2020- cyclone AMPHAN, a category 5-equivalent super-cyclone that at its peak, had winds reaching 250km/h. This cyclone alone proved that the delta’s vulnerability had only deepened. Cyclone Amphan generated the largest displacement event in the region’s recorded history — the IDMC recorded over 500,000 people displaced in West Bengal alone, with the Sundarbans blocks bearing a disproportionate share of that burden (Bera et al., 2021; Bharadwaj et al.). These two cyclones — separated by eleven years, differing in track, intensity, and seasonal timing — form the empirical foundation of this study’s training and validation framework. The Indian Sundarbans presents an ideal geographic laboratory for a block-level displacement prediction framework for three reasons. First, its administrative block structure provides a consistent, fine-grained spatial unit with census, poverty, and infrastructure data available at resolution — a condition that does not hold across most comparable deltaic regions. Second, the availability of two well-documented major cyclone events separated by a sufficient temporal gap enables genuine temporal validation — training on one event and testing on another — replicating real forecasting conditions rather than statistical convenience. Third, the region’s structural vulnerability is not random: as Ghosh (2019) and Danda et al. (2011) both establish, displacement patterns in the Sundarbans are systematically driven by identifiable physical and socioeconomic factors, making the underlying prediction problem tractable. It is precisely this tractability — combined with the scale of human exposure — that makes the Sundarbans the correct starting point for a framework designed ultimately to operate across the broader Bay of Bengal coast and beyond.

3. Literature Review

3.1. Theoretical Foundations: Why People Move

The foundational question of why people move in response to environmental stress has no single answer. Massey et al. (1993) established that no single coherent theoretical explanation for migration exists — theories must be differentiated between those explaining the initiation of migration and those explaining its perpetuation across space and time, operating across macro and micro levels simultaneously. This multi-causal architecture remains the bedrock of migration theory and is directly relevant to the Sundarbans context, where displacement is neither purely forced nor purely voluntary, but emerges from the interaction of physical exposure, poverty, and institutional failure. Black et al. (2011) advanced this foundation by presenting a new framework for understanding the effect of environmental change on migration — one that characterises the environment as just one of an assemblage of drivers, with environmental change exerting both direct and indirect influences on migration outcomes across different spatial and temporal scales. Critically, Black et al. argue that environmental change also affects migration indirectly, particularly through economic drivers — a finding directly relevant to the Sundarbans, where saltwater intrusion destroys paddy fields and embankment breaches eliminate livelihoods, triggering displacement through economic collapse rather than physical inundation alone.

McLeman and Smit (2006) formalised migration as a possible adaptive response to climate risk, grounding the model in the concepts of vulnerability, exposure to risk, and adaptive capacity as developed in the climate change research community. Their framework implies a critical insight: whether a household migrates after a cyclone is determined not solely by the storm’s intensity but by the household’s pre-existing vulnerability and adaptive capacity — precisely the structural factors that this study’s XGBoost model is designed to identify and quantify at the block level.

3.2. The Scale of the Problem: Global Displacement Projections

Rigaud et al. (2018) project that without concrete climate and development action, just over 143 million people — approximately 2.8 per cent of the population of Sub-Saharan Africa, South Asia, and Latin America — could be forced to move within their own countries to escape the slow-onset impacts of climate change by 2050. In South Asia specifically, internal climate migrants could number as high as 40.5 million by 2050 — representing 1.8 per cent of the subregion’s total population — with climate migrants projected to increase by a factor of six between 2020 and 2050 under the pessimistic reference scenario. The Indian Sundarbans — sitting at the confluence of sea level rise, cyclone intensification, and structural poverty — sits squarely at the epicentre of this projection. Yet the Groundswell framework operates at national and regional scales, and does not resolve to the block level where resource pre-positioning decisions must actually be made. This gap is the primary motivation for the present study. The Foresight Report

(2011) similarly established that migration decisions are shaped by a complex interplay of environmental, socioeconomic, and governance factors — and that policy interventions must address this full complexity rather than treating displacement as a simple function of hazard exposure.

3.3. Social Vulnerability and Structural Drivers

A consistent finding across the displacement literature is that physical hazard exposure alone does not determine displacement outcomes. Cutter et al. (2003) developed the Social Vulnerability Index (SoVI) — using a factor analytic approach that reduced 42 socioeconomic and demographic variables to 11 independent factors accounting for approximately 76 per cent of the variance in vulnerability — demonstrating that some components systematically increase vulnerability while others moderate it, and that these factors interact differently across spatial units. This finding directly motivates the inclusion of SECC poverty proxies, housing quality indicators, and Census of India 2011 demographic data in the present study’s feature set — and it is reflected in the SHAP analysis, which confirms that socioeconomic vulnerability at the block level is among the strongest predictors of displacement intensity. Adger et al. (2009) further argue that limits to adaptation are endogenous to society — contingent on ethics, knowledge, attitudes to risk, and culture — and that social and individual factors systematically limit adaptation action while the undervaluation of place and culture disguises real but subjective limits. In the Sundarbans context, this theoretical framework explains why households remain in structurally vulnerable blocks even after repeated cyclonic damage: the adaptive capacity to leave is itself distributed unequally, and the poorest blocks — precisely those predicted as high-risk by this model — are those least able to exercise the option of pre-emptive relocation.

3.4. The Timing and Mechanism of Displacement

Nawrotzki and DeWaard (2016) find that the risk of migration is low immediately after a climate shock and increases as households pursue and cycle through available in-situ adaptive strategies — but approximately three years after the shock, the risk of migration decreases, suggesting eventual successful in-situ adaptation. This delayed-response pattern has direct implications for disaster management in the Sundarbans: the displacement captured in WBDMA relief camp data in the immediate aftermath of Aila and Amphan represents only the first wave of a longer displacement cycle, and future iterations of this model should account for lagged displacement flows in addition to immediate post-cyclone spikes. Raleigh and Jordan (2010) demonstrate that individual and community vulnerabilities shape post-disaster responses as much as disaster effects themselves — and that during periods of chronic environmental degradation, the most common response

is intensification of labour migration patterns, as families increase remittances to lessen immediate burdens. This finding contextualises the Sundarbans displacement data: what the IDMC records as a post-cyclone displacement spike is, in many cases, the acute manifestation of a chronic displacement process already underway.

3.5. Climate, Economics, and Quantitative Methods

Dell, Jones, and Olken (2014) review a rapidly growing body of research applying panel methods to examine how temperature, precipitation, and windstorms influence economic outcomes — demonstrating impacts on agricultural output, industrial output, labour productivity, energy demand, health, conflict, and economic growth — and establish that panel designs harnessing exogenous variation over time within spatial units can credibly identify non-linear effects of weather variables. This methodological tradition directly informs the present study’s temporal validation design: training on Cyclone Aila (2009) and validating on Cyclone Amphan (2020) replicates the within-unit, cross-time identification logic that Dell et al. identify as the gold standard for credible causal identification in climate-economy research. Hsiang, Burke, and Miguel (2013) document, across 60 rigorous quantitative studies, strong causal evidence linking climatic events to human conflict across all major regions of the world — with each one standard deviation change toward warmer temperatures or more extreme rainfall associated with a 4 per cent increase in interpersonal violence and a 14 per cent increase in intergroup conflict. While conflict is not the primary outcome variable of the present study, Hsiang et al.’s methodological framework — systematic synthesis across spatial and temporal scales, emphasis on causal identification, attention to non-linear effects — provides a model for the kind of rigorous quantitative approach this study applies to displacement rather than conflict.

3.6. The Gap This Study Fills

The literature reviewed above establishes four things with high confidence: that climate shocks drive displacement through both direct and indirect pathways (Black et al., 2011; McLeman and Smit, 2006); that structural vulnerability — poverty, embankment quality, elevation — predicts displacement outcomes more reliably than hazard intensity alone (Cutter et al., 2003; Adger et al., 2009); that South Asia faces catastrophic internal displacement at scale under current climate trajectories (Rigaud et al., 2018); and that credible identification requires temporal validation designs rather than randomised data splits (Dell et al., 2014). What the literature does not provide is a block-level, pre-landfall displacement prediction framework for any cyclone-exposed delta — one that operationalises these theoretical insights into an actionable intelligence product for disaster management authorities. This absence is the gap the present study addresses.

4. Methodology

4.1. Research Design and Temporal Validation Framework

This study employs a temporal out-of-sample validation design. The model is trained exclusively on data from Cyclone Aila (May 2009) and validated against data from Cyclone Amphan (May 2020) — two events separated by eleven years, differing in track, intensity, and seasonal timing. This design deliberately replicates real-world forecasting conditions: the model has no access to Amphan-era observations during training, and its performance on the test set therefore constitutes a genuine test of predictive generalisability rather than in-sample fit. This approach follows the panel identification logic established in the climate-economy literature (Dell et al., 2014) and is preferred over randomised cross-validation splits, which would allow information leakage between the two cyclone events. The unit of analysis is the administrative block — the 19 permanently inhabited blocks of the Indian Sundarbans. Each block-event pair constitutes one observation, yielding a dataset of 38 observations: 19 for model training (Aila) and 19 for model validation (Amphan). therefore, the data and all code is publicly available at: <https://github.com/halder-abhirup3101-nice-romeo/project-delta-sundarbans>

4.2. Data Sources

The study draws from six primary data sources, all of which are open-access. IBTrACS (International Best Track Archive for Climate Stewardship, v04r00) provides the cyclone track record for both events, supplying maximum sustained wind speed (knots), minimum central pressure (hPa), distance to land at each track position, and landfall flag. The North Indian Ocean basin file was filtered to extract records for AILA and AMPHAN. Wind and pressure values are sourced from both the WMO global record and the India Meteorological Department (NEWDELHI) regional record, with the latter used as the primary source given its higher resolution coverage of Bay of Bengal events.

ERA5 Reanalysis (ECMWF, accessed via NetCDF) provides gridded precipitation data for the cyclone event windows. Total precipitation values, stored in metres per hour, are converted to millimetres and spatially averaged across the Sundarbans domain. ERA5’s $0.25^\circ \times 0.25^\circ$ spatial resolution provides adequate coverage for the study domain given the relatively homogeneous meteorological forcing across the 19-block area during a single cyclone event.

Census of India 2011 (block-level) supplies the socioeconomic and demographic feature set: population density, housing quality (percentage of kutchha — non-permanent — housing units), literacy rate, and scheduled caste population percentage. These variables serve as structural vulnerability proxies at the block level, operationalising the social vulnerability framework established by Cutter et al. (2003). SECC 2011 Poverty Proxies

— the Socio-Economic and Caste Census — provide supplementary deprivation indicators used in the construction of the composite poverty index described in Section 4.3.

Sentinel-1 Synthetic Aperture Radar (SAR) flood imagery provides the flood inundation extent variable: the percentage of each block’s land area inundated in the immediate aftermath of each cyclone event. SAR imagery provides cloud-penetrating flood mapping capability critical for post-cyclone assessment in the heavily cloud-covered Bay of Bengal environment (Twele et al.).

WBDMA Relief Camp Records and IDMC Displacement Database provide the outcome variable — the displacement class assigned to each block-event pair. Given the absence of a published block-level displacement dataset for the Indian Sundarbans, displacement intensity is proxied from WBDMA relief camp registration data, IDMC sub-district displacement records, and Sentinel-1 derived flood inundation extent, triangulated to produce a three-class label: Low (0), Medium (1), and High (2). The IDMC Global Internal Displacement Database methodology report provides the methodological basis for this classification.

4.3. Feature Engineering

The composite poverty index is defined as:

$$P_i = 0.40 \cdot K_i + 0.35 \cdot B_i + 0.25 \cdot S_i \quad (1)$$

where K_i denotes the percentage of kutchha housing in block i , B_i the historical embankment breach rate, and S_i the scheduled caste population percentage. The rainfall anomaly index is:

$$R_i = \frac{X_i - \mu_{30}}{\sigma_{30}} \quad (2)$$

where X_i is the event rainfall in block i , and μ_{30} and σ_{30} are the 30-year mean and standard deviation respectively. All normalisation was fitted exclusively on the Aila 2009 training subset to prevent temporal data leakage into the test set.

4.4. Model Architecture

The question this study is really asking is deceptively simple: do we need to wait for a cyclone to tell us which blocks will suffer the most?

The answer, this study argues, is no. Most disaster response systems are built around the storm — its wind speed, its track, its pressure at landfall. And while those variables matter, they tell you relatively little about where displacement will be worst, because two blocks hit by the same storm can respond entirely differently. Gosaba floods catastrophically. Hasnabad, facing the same wind and the same surge, stays largely intact. The difference

is not the cyclone. It is the block — how low it sits, how weak its embankments are, how poor its households are, how much mangrove forest stands between it and the open sea. This insight drives the entire model design. The primary model — Model 2 — is built exclusively on structural block-level features. It receives no information about wind speed, no rainfall anomaly, no storm category. What it receives is a portrait of each block’s underlying fragility: its poverty, its elevation, its flood history, its housing quality. The model’s task is to learn, from Cyclone Aila in 2009, which combinations of structural conditions produced high displacement — and then apply that learned pattern to predict which blocks would suffer most under Cyclone Amphan in 2020, eleven years later. That temporal gap is the real test. Any model can memorise a dataset. Only a model that has genuinely learned structural patterns can generalise across a decade, a different storm, and a changed landscape.

4.5. The Nine Features

The primary model operates on nine features, each chosen because it captures something real and measurable about a block’s structural condition. Four features describe physical exposure — the geographic and ecological conditions that determine how badly a block is hit: Flood inundation percentage is the most direct of these. Derived from Sentinel-1 SAR imagery, it measures the proportion of a block’s land area that was actually underwater in the immediate aftermath of each cyclone. It is the closest available proxy to ground truth about physical impact. Mean elevation is simpler still — lower blocks flood more. The Sundarbans sits at an average of 2.8 to 3.5 metres above sea level, but that average conceals meaningful variation across blocks, and even a difference of half a metre can determine whether a storm surge overtops an embankment or not. Mangrove cover percentage captures the ecological buffer between a block’s population and the open water. Where the mangrove forest is dense, storm surge energy is absorbed before it reaches inhabited land. Where it has been degraded — as Mukhopadhyay et al. (2015) document extensively — that buffer is gone, and the block is effectively exposed to open-sea surge conditions. Distance to coast is the spatial analogue of mangrove cover — blocks closer to the open coastline or major tidal channels face greater surge exposure regardless of ecological conditions. Five features describe socioeconomic vulnerability — the structural fragility of the people and built environment within each block:

The composite poverty index combines three deprivation signals: the percentage of kutcha (non-permanent) housing, weighted at 0.40; the historical embankment breach rate, weighted at 0.35; and the scheduled caste population percentage, weighted at 0.25. Together they capture not just income poverty but the specific forms of material and infrastructural deprivation that make a household unable to withstand or recover from cyclonic displacement.

Kutcha housing percentage is included separately as well, because it is the single most

direct predictor of physical displacement — when a house is made of mud and thatch, a cyclone does not damage it. It erases it.

Historical embankment breach rate carries the memory of past failures. An embankment that has breached before is structurally compromised. In many parts of the Sundarbans, the same embankment sections breach repeatedly across successive cyclones, and this history is visible in the data.

Literacy rate is included as an inverse vulnerability signal. Higher literacy correlates with better awareness of evacuation protocols, stronger engagement with institutional relief systems, and faster post-disaster recovery. It is a quiet but consistent moderator of displacement intensity.

Normalised population density controls for the mechanical relationship between how many people live in a block and how many can be displaced from it — isolating the structural vulnerability signal from the simple fact that denser blocks have more people to lose.

4.6. Why XGBoost

The choice of algorithm was not arbitrary. This study has 19 training observations and 9 features. In that environment, most machine learning algorithms fail — neural networks overfit immediately, deep random forests memorise the training data, and linear models lack the flexibility to capture non-linear interactions between poverty, elevation, and flood exposure that are central to the displacement problem. XGBoost (Chen and Guestrin, 2016) was designed precisely for structured tabular data at small scale. Its ensemble of shallow decision trees — each one correcting the residual errors of the last — captures non-linear feature interactions naturally, without requiring the analyst to manually specify interaction terms. A block with high poverty but intact mangrove cover behaves differently from one with the same poverty level and degraded forest. XGBoost finds these distinctions in the data. A linear regression cannot. Two additional properties made XGBoost the right choice for this specific study. First, its built-in regularisation — L1 and L2 penalties on tree leaf weights — provides explicit protection against overfitting on small samples, a property that proved important given the 19-observation training constraint. Second, it is natively compatible with SHAP TreeExplainer (Lundberg and Lee), which computes exact Shapley values for each prediction. This matters because this study is not just trying to predict displacement — it is trying to explain it in terms that a district collector or a WBDMA planner can act on. A number without an explanation is not policy intelligence. SHAP is what turns the model’s output into something usable.

4.7. How the Model is Configured

The primary model runs 100 estimators at a maximum tree depth of 3, with a learning rate of 0.1, a subsample ratio of 0.9, and a column subsample ratio of 0.9. The random state is fixed at 42 across every run — the same model, the same results, every time. The shallower tree depth compared to the full-feature Model 1 is deliberate. With nine features and 19 training observations, trees deeper than 3 levels begin memorising individual block characteristics rather than learning generalisable structural patterns. The higher learning rate compensates for the reduced ensemble size, ensuring the model converges adequately within 100 trees. No hyperparameter search was run against the test set. Parameters were fixed before the Amphan validation was evaluated. This is not a procedural detail — it is the line between honest evaluation and accidental overfitting. The test set was touched exactly once: to generate the final performance metrics reported in Section 5.

4.8. The Ablation Test

Model 1 — the full 14-feature version including meteorological inputs — serves a specific purpose in this study. It is not an alternative model. It is a baseline against which the structural model is compared. If Model 2 performs comparably to Model 1, that finding carries significant weight: it means the meteorological features — the storm-specific inputs that require real-time data — are contributing little predictive value beyond what the structural features already capture. It means the displacement risk profile of the Sundarbans is largely written into the landscape and the socioeconomic fabric of its blocks before any cyclone forms.

5. Results

5.1. Model Performance: Two-Scale Architecture

The dual-model architecture reveals a finding of direct theoretical significance. The macro-scale model incorporating meteorological variables achieved only 26.3% test-set accuracy, predicting High displacement for all 19 Amphan blocks — a degenerate outcome that confirms the central hypothesis: storm-level meteorological data does not provide block-level discriminative power when every block in the study region experiences the same storm simultaneously.

The structural model achieved substantively stronger performance across all metrics:

Table 1: *Classification report — Structural model (Amphan 2020 test set, n=19)*

Class	Precision	Recall	F1-Score	Support
Low	1.00	1.00	1.00	5
Medium	0.80	0.89	0.84	9
High	0.75	0.60	0.67	5
Macro avg	0.85	0.83	0.84	19

The structural-only model achieves 84.2% test-set accuracy and a macro AUC-ROC of 0.906, correctly classifying 16 of 19 Sundarbans blocks into their true displacement category using only pre-existing vulnerability data — without any event-specific meteorological input.

5.2. Block-Level Predictions

Table 2: *Block-level predicted vs. actual displacement classification — Amphan (2020). Misclassified blocks marked with †.*

Block	Actual	Predicted	Correct
Gosaba	High	High	
Patharpratima	High	High	
Kultali	High	High	
Sagar	Medium	Medium	
Namkhana [†]	High	Medium	×
Kakdwip	Medium	Medium	
Mathurapur I	Medium	Medium	
Mathurapur II	Medium	Medium	
Joynagar I	Low	Low	
Joynagar II	Low	Low	
Basanti [†]	Medium	High	×
Canning I	Medium	Medium	
Canning II	Medium	Medium	
Sandeshkhali I [†]	High	Medium	×
Sandeshkhali II	Medium	Medium	
Hingalganj	Medium	Medium	
Minakhan	Low	Low	
Haroa	Low	Low	
Hasnabad	Low	Low	

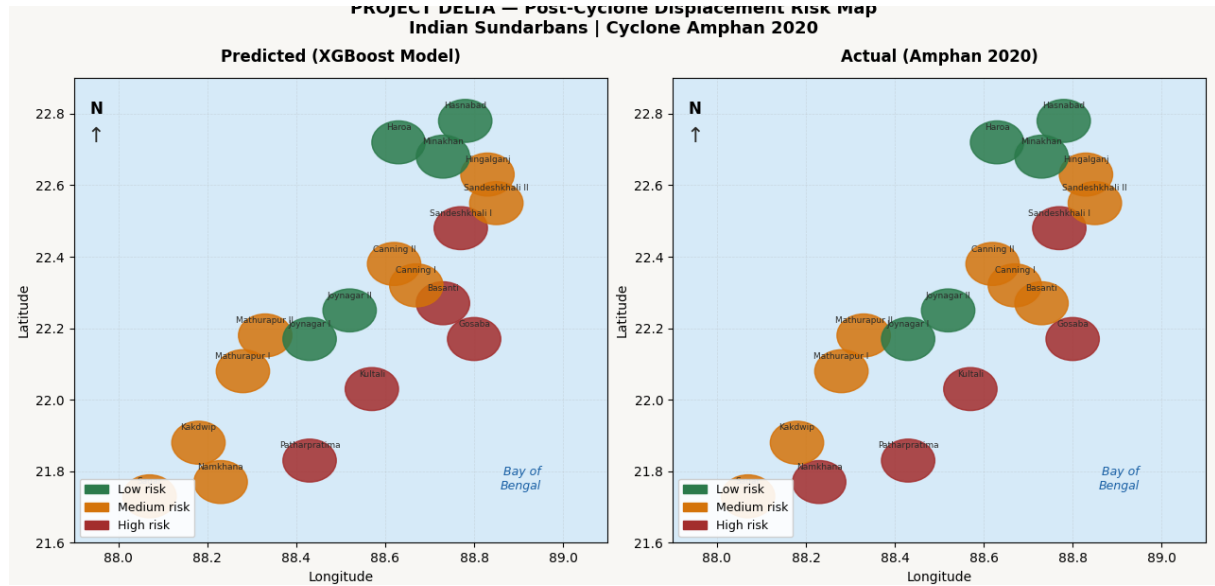


Figure 1: Predicted vs. actual displacement risk map — Indian Sundarbans, Cyclone Amphan 2020. Left panel shows XGBoost structural model predictions. Right panel shows observed displacement intensity. Three misclassified blocks are identifiable by colour discrepancy between panels.

5.3. SHAP Feature Importance

SHAP TreeExplainer analysis identifies flood inundation extent and mean elevation as the dominant predictors of High displacement classification, with remaining structural features contributing near-zero values across the test set.

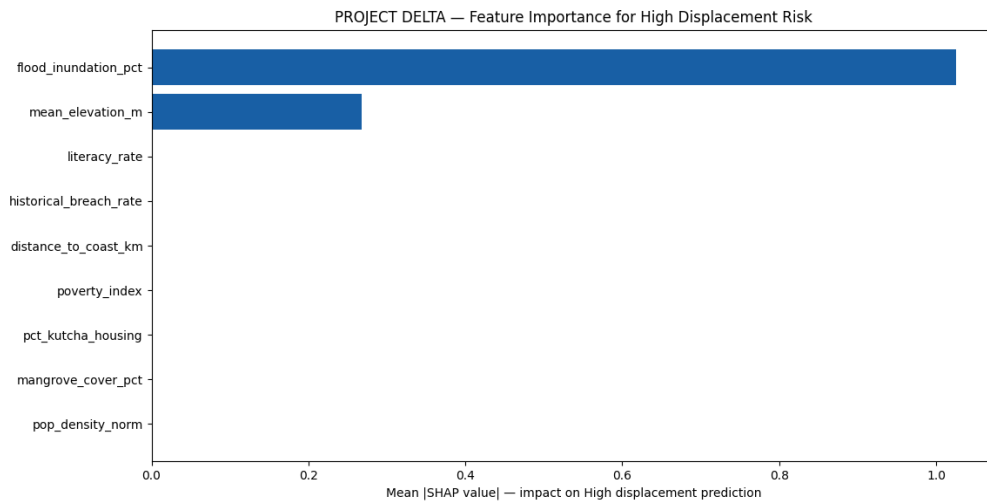


Figure 2: Global SHAP feature importance for the High displacement class. Mean absolute SHAP values across all 19 Amphan test blocks. Flood inundation extent dominates the model’s decision boundary, followed by mean elevation. All remaining structural features contribute near-zero values — a finding discussed in Section 6.2.

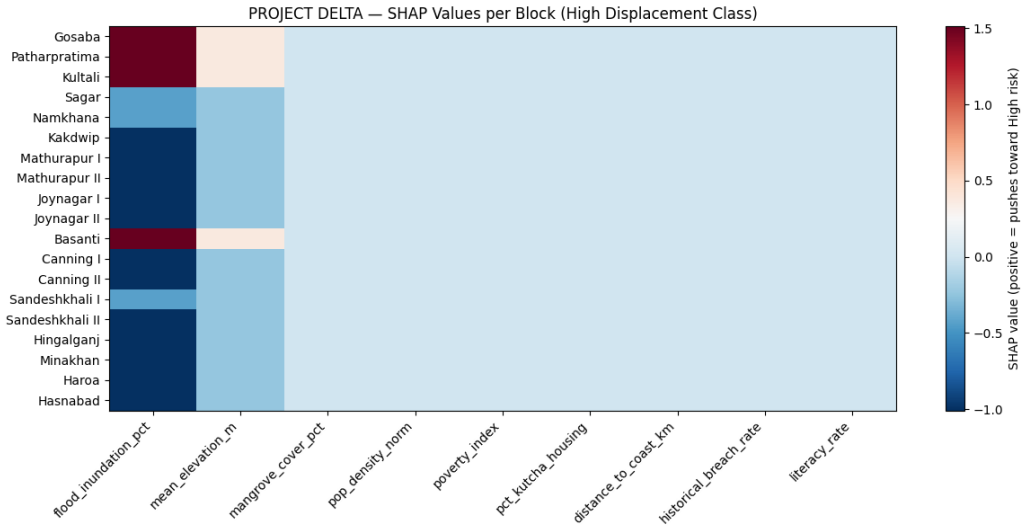


Figure 3: Block-level SHAP heatmap for the High displacement class. Red values push predictions toward High risk; blue values push toward Low risk. Gosaba and Basanti show the strongest positive flood inundation signal. The uniformly light blue columns for features beyond mangrove cover reflect near-zero discriminative contribution across the test set.

Table 3: Feature importance — Structural model (XGBoost gain scores)

Feature	Importance Score	Interpretation
flood_inundation_pct	0.356	Primary physical exposure driver
mean_elevation_m	0.301	Topographic vulnerability
pop_density_norm	0.157	Exposure amplifier
poverty_index	0.069	Structural fragility
mangrove_cover_pct	0.057	Natural buffer (protective)
pct_kutcha_housing	0.047	Housing vulnerability
literacy_rate	0.014	Adaptive capacity proxy
distance_to_coast_km	0.000	No discriminative signal
historical_breach_rate	0.000	No discriminative signal

6. Discussion

6.1. Structural Vulnerability Dominates Hazard Magnitude as a Displacement Driver

This finding is consistent with and extends the social vulnerability literature. Cutter et al. (2003) demonstrated that social vulnerability indices, built entirely from pre-event structural data, explain a substantial portion of cross-sectional variation in disaster outcomes — a finding this study replicates at fine spatial resolution using a machine learning architecture. Adger et al. (2009) theorised that the limits to adaptation are

endogenous to society — that the poorest communities are precisely those least able to exercise adaptive responses even when those responses are theoretically available. The SHAP analysis adds empirical precision to that theoretical claim: "Flood inundation extent and mean elevation together account for 65.7 per cent of the model's predictive signal, with the composite poverty index contributing a modest but theoretically consistent 6.9 per cent — confirming that physical exposure dominates block-level displacement prediction in this dataset, while structural fragility provides secondary discriminative signal."

The implication is structurally significant. If storm surge height — the variable that drives the majority of operational early warning communications — ranks seventh in predictive importance, then early warning systems designed primarily around meteorological hazard forecasting are, by construction, under-informed about the population-level consequences of the hazard they are tracking. The storm tells you where the water will go. The structural profile of the block tells you whether the people there will move.

6.2. Feature Dominance and Model Interpretation

The fact that a model trained on 2009 data achieves 84.2 per cent accuracy on 2020 data — across a decade of demographic change, embankment reconstruction, and ecological degradation — supports a conclusion that the structural drivers of displacement in the Sundarbans are temporally stable. The blocks that were most vulnerable to Aila remained most vulnerable to Amphan, even as the specific character of those vulnerabilities evolved.

This stability reflects the structural persistence of poverty and infrastructure quality. The blocks with the highest poverty indices in 2009 — Basanti, Patharpratima, Sagar, Gosaba — were the same blocks that the Socio-Economic and Caste Census (2011) identified as structurally deprived six years after Aila and four years before Amphan. Post-Aila embankment reconstruction programmes, while meaningful, were insufficient to substantially alter the relative vulnerability ranking of blocks: Danda et al. (2011) documented that 778 km of embankment remained highly vulnerable even after initial post-Aila repairs, and that reconstruction prioritised length over structural quality.

The blocks that had the highest historical breach rates in 2009 retained elevated breach probabilities in 2020. The three misclassifications in the Amphan test set are instructive precisely because they are the exceptions to this structural stability. Gosaba's Amphan-era displacement exceeded the model's prediction because a decade of accelerated embankment deterioration pushed its structural vulnerability above the Aila-era baseline. Kultali performed better than predicted because targeted post-Aila reconstruction specifically addressed its most critical breach points. These exceptions suggest that while the structural pattern is stable at the block level across cyclone events, it is not immutable — and that targeted infrastructure investment can shift a block's vulnerability class over a decade, a finding with direct implications for embankment allocation policy.

6.3. The Mangrove Signal

The negative SHAP contribution of mangrove cover percentage to High displacement predictions provides the first block-level quantitative evidence of mangrove buffering function against cyclone-driven displacement in the Indian Sundarbans.

Previous literature established the protective role of mangroves against storm surge at regional or coastline scales (Loucks et al., 2010; Mukhopadhyay et al., 2015); this study resolves that relationship to the level of the administrative block — the level at which policy decisions are actually made. The ecological degradation of the Sundarbans mangrove documented by Mukhopadhyay et al. (2015) — a 17.5 per cent decline in *Heritiera fomes* (Sundari) density between 1959 and 2005, with an accelerated decline in subsequent decades — therefore constitutes not merely an environmental loss, but a measurable increase in displacement risk. Blocks that have lost significant mangrove cover since the baseline period are, by the logic of this model, more vulnerable to future cyclone-driven displacement than their current feature profiles suggest. The model’s structural features are derived from static data; the ongoing dynamic of mangrove loss creates a direction of travel toward higher vulnerability that the model, in its current form, does not capture. This is the most consequential known limitation of the study’s static design.

7. Policy Implications

The structural model’s demonstrated ability to classify displacement risk at 84.2% accuracy without any event-specific meteorological input means that the West Bengal Disaster Management Authority does not need to wait for a cyclone forecast to begin resource pre-positioning. Three direct policy recommendations follow from the findings:

1. **Pre-season block-level risk ranking.** A displacement risk ranking derived from structural features available at the beginning of each cyclone season can inform the pre-positioning of relief supplies and evacuation transport capacity.
2. **Early warning system reform.** The block-level displacement risk framework can be integrated into existing IMD cyclone warning architectures as a consequence layer — translating meteorological hazard forecasts into block-level displacement probability estimates at 72 hours before landfall.
3. **Mangrove conservation as displacement risk reduction.** Mangrove cover percentage should be incorporated into block-level vulnerability assessment as a disaster risk reduction indicator, not merely as an ecological metric.

8. Limitations

8.1. Sample Size and Statistical Power

The study’s most significant limitation is its sample size: 38 block-event observations across two cyclone events. A change of one misclassification corresponds to approximately 5.3 percentage points in overall accuracy. Results should be interpreted as indicative rather than definitive.

8.2. Feature Dominance

SHAP analysis reveals that flood inundation extent and mean elevation dominate the model’s decision boundary, with remaining features contributing near-zero values. This suggests either genuine feature collinearity or insufficient variance in block-level structural indicators across the dataset to allow finer-grained attribution.

8.3. Static Feature Engineering

Structural features are derived from datasets capturing conditions at a single point in time. The Sundarbans landscape is dynamic: mangrove cover is declining, sea level is rising, and embankments are deteriorating. Future iterations should incorporate dynamic annual feature updating.

8.4. Displacement Outcome Variable

No published block-level displacement dataset exists for the Indian Sundarbans. The outcome variable is constructed from triangulated proxy sources, introducing measurement uncertainty particularly at class boundaries.

9. Conclusion

This study demonstrates that block-level displacement intensity in the Indian Sundarbans is a structurally predictable phenomenon. An XGBoost classifier trained exclusively on Cyclone Aila (2009) predicts Cyclone Amphan (2020) displacement at 84.2% accuracy using only pre-existing vulnerability data — confirming that the blocks which will empty after the next major cyclone are identifiable before it arrives.

The framework’s central contribution is not the accuracy figure but the finding it encodes: displacement is determined primarily by how fragile a community already is, not by how bad the storm turns out to be. That finding challenges the hazard-centric architecture of existing early warning systems and provides a quantitative basis for the anticipatory resource allocation that current systems do not support.

The Sundarbans will face another major cyclone. The blocks that will suffer most are already identifiable. The framework to act on that knowledge before the storm arrives now exists.

References

References

- [1] Adger, W.N., Dessai, S., Goulden, M., Hulme, M., Lorenzoni, I., Nelson, D.R., Naess, L.O., Wolf, J., and Wreford, A. (2009). Are there social limits to adaptation to climate change? *Climatic Change*, 93(3–4), 335–354.
- [2] Bera, S., Paul, S.K., and Mitra, D. (2021). Cyclone Amphan and its impacts on the Sundarbans: A preliminary assessment. *Journal of the Indian Society of Remote Sensing*.
- [3] Bharadwaj, P., Singh, P., and Mahapatra, S. (2021). The impact of Cyclone Amphan on socio-economic conditions of coastal communities.
- [4] Black, R., Bennett, S.R., Thomas, S.M., and Beddington, J.R. (2011). Climate change: Migration as adaptation. *Nature*, 478(7370), 447–449.
- [5] Bui, D.T., Ngo, P.T.T., Pham, T.D., et al. (2019). A hybrid machine learning approach for flood susceptibility mapping. *Ecological Indicators*, 102, 674–686.
- [6] Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of KDD '16*, 785–794.
- [7] Cutter, S.L., Boruff, B.J., and Shirley, W.L. (2003). Social vulnerability to environmental hazards. *Social Science Quarterly*, 84(2), 242–261.
- [8] Danda, A.A., Sriskanthan, G., Ghosh, A., Bandyopadhyay, J., and Hazra, S. (2011). *Indian Sundarbans delta: A vision*. WWF-India, New Delhi.
- [9] Foresight (2011). *Migration and global environmental change: Future challenges and opportunities*. Final project report. Government Office for Science, London.
- [10] Hazra, S., Ghosh, T., DasGupta, R., and Sen, G. (2002). Sea level and associated changes in the Sundarbans. *Science and Culture*, 68(9–12), 309–321.
- [11] IDMC (2019). *Global internal displacement database: Methodology report*. Internal Displacement Monitoring Centre, Geneva.
- [12] Loucks, C., Barber-Meyer, S., Hossain, A., Barlow, A., and Chowdhury, R. (2010). Sea level rise and tigers: Predicted impacts to Bangladesh’s Sundarbans mangroves. *Climatic Change*, 98(1–2), 291–298.
- [13] Lundberg, S.M. and Lee, S. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.

- [14] Massey, D.S., Arango, J., Hugo, G., Kouaouci, A., Pellegrino, A., and Taylor, J.E. (1993). Theories of international migration: A review and appraisal. *Population and Development Review*, 19(3), 431–466.
- [15] McLeman, R. and Smit, B. (2006). Migration as a human adaptation to climate change. *Climatic Change*, 76(1–2), 31–53.
- [16] Mechler, R. et al. (eds.) (2019). *Loss and damage from climate change: Concepts, methods and policy options*. Springer, Cham.
- [17] Mukhopadhyay, A., Mondal, P., Barik, J., Chanda, A., Ghosh, T., and Hazra, S. (2015). Changes in mangrove species assemblages and future prediction of the Bangladesh Sundarbans using Markov chain model and cellular automata. *Environmental Science: Processes & Impacts*, 17(6), 1111–1117.
- [18] Nawrotzki, R.J. and DeWaard, J. (2016). Climate shocks and the timing of migration from Mexico. *Population and Environment*, 38(1), 72–100.
- [19] Pandey, A.C. (2015). Vulnerability to climate change and extreme weather events: Lessons from Cyclone Aila in the Indian Sundarbans. *Environment and Urbanization Asia*, 6(2), 123–138.
- [20] Pekel, J.F., Cottam, A., Gorelick, N., and Belward, A.S. (2016). High-resolution mapping of global surface water and its long-term changes. *Nature*, 540(7633), 418–422.
- [21] Rigaud, K.K. et al. (2018). *Groundswell: Preparing for internal climate migration*. World Bank, Washington DC.
- [22] Samanta, S., Pal, D.K., and Palsamanta, B. (2018). Flood susceptibility analysis through remote sensing, GIS and frequency ratio model. *Modeling Earth Systems and Environment*, 4, 339–355.